**25901656** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

## SECOND SEMESTER

## Discipline Specific Core Course (DSC) - MG2DSCECT100 - ESSENTIAL CONCEPTS IN DIGITAL ELECTRONICS

(2024 ADMISSION ONWARDS)

**Duration: 1 Hour 30 Minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which base does the hexadecimal number system use? [K] / [CO1]

   a). 10 b). 8

   c). 16 d). 2

2. To create an XOR gate using only NAND gates, at least how many gates are required? [K] / [CO2]

   a). 2 b). 3

   c). 4 d). 5

3. Find the 2's complement of 010100 [U] / [CO1]

   a). 101100 b). 101101

   c). 101110 d). 110011

4. Binary subtraction of 10100 - 1101 equals? [U] / [CO1]

a). 1101 b). 1111

c). 111 d). 1010

5. Which of the following correctly represents De Morgan's theorem in Boolean algebra? [U] / [CO2]

a). $\overline{A.B} = \overline{A} + \overline{B}$ b). $'\overline{A+B} = \overline{A}.\overline{B}$

c). Both A and B d). None of the Above

6. Determine the decimal value of $(10011001)_2$ in the sign-magnitude form. [U] / [CO1]

a). -25 b). +25

c). -22 d). -24

7. What is the binary representation of the decimal number 9? [K] / [CO1]

a). 1000 b). 1001 c). 1010 d). 1111

8. An example of product- of -sums expression is [U] / [CO2]

a). AB+AC b). (A+B)(A+B+C)

c). A(B+C)+AC d). both answers(a) and(b)

9. Which logic operation does the Carry output of a Half Adder follow? [U] / [CO3]

a). AND b). XOR

c). NOR d). XOR

10. In a 4-bit SISO shift register, if the initial data is 1101 and 0 is shifted in, what will the final output be after 4 clock pulses? [K] / [CO3]

a). 1100 b). b)0000 c). 1111 d). 1001

**[1x10 = 10]**

## Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Differentiate between analog and digital signals with examples. [U] / [CO1]

12. Use a K-map to find the minimum POS form for the expression $(A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(A+\bar{B}+\bar{C})+(\bar{A}+\bar{B}+C)$ [U] / [CO2]

13. Explain the working principle of a D flip-flop. How does it differ from an SR flip-flop? [K] / [CO3]

14. How can you implement an OR gate using only NAND gates? [K] / [CO2]

15. How does the clock signal affect the operation of an SR flip-flop? [K] / [CO3]

16. Convert the following decimal numbers into binary form: (a) 103 (b) 86 [U] / [CO1]

**[5x4 = 20]**

## Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Define the following logic gates i)AND gate ii) OR gate iii) NOT gate iv) XOR gate For each gate, provide the truth table, logic symbol, and Boolean expression. [U] / [CO2]

18. Explain 4:1 multiplexer circuit in detail? [U] / [CO3]

19. State and describe the rules and laws of Boolean Algebra with examples [U] / [CO2]

20. Compare and contrast the binary and hexadecimal number systems in terms of their structure, representation, and usage. [K] / [CO1]

**[10x2 = 20]**